

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP- 1st Year 1st Term Examination, 2016

CH 1117
(Environmental Chemistry)

Full Marks: 210

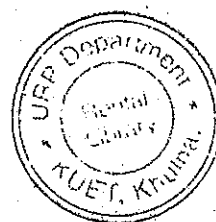
Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) An S_N2 reaction proceeds with complete stereochemical inversion – Explain. (10)
(b) 3° Carbocation is more stable than 1° Carbocation. Why? (07)
(c) Discuss the stereochemistry of S_N1 reaction mechanism. (10)
(d) What is activation energy of a reaction? (08)
2. (a) Define polymer and macromolecules with examples. (08)
(b) Discuss the free radical mechanism of polymerization process. (12)
(c) What is environmental chemistry? Discuss the social relevance. (10)
(d) Write down the composition of atmosphere. (05)
3. (a) Write down the characteristic properties of oligotrophic and eutrophic lake. (08)
(b) Define industrial hazards. Write down the different types of chemical hazards with example. (07)
(c) Write down the name of different water pollutants. What are the non-point sources of water pollution? (10)
(d) Describe the different segments of environment in short. (10)
4. (a) Define corrosion and erosion giving examples. (06)
(b) Write down the characteristics of wastewater. Discuss briefly the treatment process of domestic wastewater. (10)
(c) Discuss the electro-chemical mechanism of wet corrosion. (09)
(d) Describe the role of pH, dissolved Oxygen and salt concentration on the rate of corrosion. (10)

Section – B



5. (a) Define different types of Hydrogen bonding with examples. (08)
(b) Boiling point of H_2O is $100^\circ C$ whereas it is $-60^\circ C$ of H_2S . Explain chemically. (07)
(c) Discuss the nature of hybridization, shape and bond angle of CH_3CH_3 , H_2O and NH_3 molecules. (12)
(d) 50g liquid water shows higher density than that of 50g ice (solid). Explain chemically. (08)
6. (a) What is metallic bonding? Explain. (07)
(b) Outline the electron sea model of metallic bonding. (12)
(c) Copper wire shows high conductance. Explain. (08)
(d) Explain the bond order and magnetic nature of N_2 and N_2^+ species considering the concept of MOT. (08)

7. (a) What is radioactivity? Describe the properties of the radiation obtained from radium? What information has this knowledge contributed to the mechanism of radioactivity? (10)
- (b) What is group displacement law? (06)
- (c) Outline the uranium decay series. (07)
- (d) What do you mean by mass defect and binding energy? Why does iron exist as the largest amount among metals in this earth? (12)
8. (a) Explain what is meant by molarity and molality of solution. Which of these two is temperature dependent? (10)
- (b) The amount of ^{14}C in a dead deer is found to be one-tenth of its amount in a fresh alive deer in Iceland. Calculate the age of the dead deer [Given, $t_{1/2}$ of ^{14}C is 7777 years]. (08)
- (c) Discuss the Lewis theory of acids and bases. (07)
- (d) State and explain the Raoult's Law. Predict about ideal and non-ideal nature of solutions with respect to this law. (10)



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Examination, 2016
Hum 1117
(Functional English)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Make W/H questions from the underlined parts of the following answers: (14)
 - i. Dhaka is the capital of Bangladesh.
 - ii. Khulna is famous for Sundarban.
 - iii. He earns 40 thousand taka per month.
 - iv. We have achieved our independence in 1971.
 - v. His father has been ill for several days.
 - vi. I read in KUET.
 - vii. This pen costs 50 taka.
- (b) Make sentences using the words as directed: (12)
Close (as Noun); Down (as Adjective); Down (as Adverb); But (as Pronoun); But (as Conjunction); Chair (as Verb).
- (c) Make new words with each of the following prefixes and suffixes and use the new words in a sentence. (09)
Im_____, Em_____, Equi_____, _____en, _____ly, _____ise.
2. (a) Make sentence with the following modals as asked in brackets: (14)
 - i. Ought to (To express advisability).
 - ii. Should (To express natural inference).
 - iii. Should (To express past duty which was not discharged).
 - iv. Would (To express polite request).
 - v. Must (To express compulsion).
 - vi. Need (To express advisability).
 - vii. Be + going + to (To express intention).
- (b) Make sentences on the following structures using the verbs given in brackets. (12)
 - i. Subj. + Intransitive verb + adverbial + extension. (Work as Verb)
 - ii. Subj. + Linking verb + adjective complement + extension. (Turn as Verb)
 - iii. Subj. + Linking verb + non complement + extension. (Become as Verb)
 - iv. Subj. + Transitive verb + infinitive as object. (Want as Verb)
 - v. Subj. + Transitive verb + gerund as object. (Finish as Verb)
 - vi. Subj. + Transitive verb + object + adj. complement. (Declare as Verb)
- (c) Fill in the gaps of the following sentences : (09)
 - i. His table is heavier than _____.
 - ii. Neither Biva nor her friends _____ done their task.
 - iii. Either of the two boys _____ talent.
 - iv. There are two pens. Neither _____ good.
 - v. The station of Kampanigong is not as big as _____ of Khulna.
 - vi. He was a doctor and we believed him to be as _____.
3. (a) Transform the sentences as directed. (14)
 - i. I am sure that he will succeed. (Simple)
 - ii. Why did not they give the proof? (Passive)
 - iii. No one will deny that he is brave. (Affirmative)
 - iv. He is, at least, as wise as any other boy. (Comparative)
 - v. His honesty is known to all. (Complex)
 - vi. Your play is death to us. (Complex)
 - vii. He escaped unhurt. (Interrogative)
- (b) Differentiate direct and indirect object with example. Define intransitive verb with example. (12)



- (c) Make one antonym and one synonym with each of the following words and use them in sentences. (09)
Innocent; Heroic; Lovely; Indecent.

4. (a) Correct the following sentences: (14)
i. I read ittefaque daily.
ii. He lives in flat.
iii. He lives in the boarding house.
iv. I have been to Natore, Bogra, Khulna and others.
v. You what say covers the whole of socialism.
vi. Not only he participate but I in the meeting.
vii. He as ill he can't attend in the meeting.
- (b) Identify the parts of speech of the underlined words of the following sentences. (12)
i. Who is that boy?
ii. The book that I bought yesterday has been stolen.
iii. That is just what I want.
iv. I am so glad you have passed.
v. He was ill, so he could not go.
vi. Since you are ill, you could not go.
- (c) Define gerund, present participle and infinitive with example. (09)

Section – B

5. (a) Read the following passage carefully and answer the following questions that follow: (20)
- The concept of social mobilization is of recent origin although it began to emerge as a sporadic activity during the late seventies and early eighties; the development thinkers and workers realized that the government machinery and the functionaries alone are not adequate to achieve development of the people specially those who are underprivileged, deprived or disadvantaged. Social mobilization is based on the assumption that no matter how worthy a cause, little progress can be made in achieving it until public support and commitment is built up and diverse sectors of the society become actively involved in the process of expected change. People who have moral or political authority or power to influence, to help achieve a specific goal are identified as target groups for social mobilization. Most countries for social mobilization begin with political leaders, decision makers and opinion leaders. Programme managers and providers of services, such as health workers, teachers, extension workers and cooperative agents from target groups also. In many countries government officials both at central and local levels are mobilized. These include ministries and institutions dealing with education, information, social affairs, small scale industries, planning and finance, etc. In the non-government sector, target groups for mobilization are usually voluntary organizations, private sector, media producers and advertisers, local, traditional and religious institutions. Finally it can be remarked that social mobilization brings the world in such a state of development in this era.
- i. Why is social mobilization need in a country?
 - ii. What should we do to have an expected change in society?
 - iii. For social mobilization who are chosen?
 - iv. In a non-government sector who are chosen for social mobilization?
- (b) Make a précis of the above written passage (Q.5.a) with a suitable title. (15)
6. (a) Amplify the idea contained in the following statement. (20)
Time is money.
- (b) Write a report on medical centre of KUET. (Around 800 words). (15)
7. (a) Amplify the idea "Smoking or health: The choice is yours". (Around 1200 words). (20)
- (b) Write a short note on our National Mourning Day. (15)
8. Write a free composition on one of the followings: (Around 2000 words). (35)
- (a) The role of education in nation building.
 - (b) Development and national unity.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2016

URP 1113
(Fundamentals of Planning Process)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Write down your understanding about Urban and Regional Planning? (05)
(b) What are the ways to define urban planning as per UN Habitat book named *Urban Planning for City Leaders*? (10)
(c) "VMOSA (Vision, Mission, Objectives, Strategies and Action Plans) is a practical planning process that can be used by any community, organization or initiative" – Briefly explain to justify the statement. (10)
(d) The life cycle of urban development contains several phases. Briefly explain the phases with pertinent examples. (10)
2. (a) What are the conditions to be an urban area (Pourashava) in Bangladesh? Discuss various types of urban areas of Bangladesh. (10)
(b) "Planning has three dimensions in terms of Space, Sector and time". Briefly explain with relevant examples in the context of Bangladesh. (10)
(c) Conceptual Planning process of Systematic Planning approach is narrated well by Brian McLoughlin, George Chadwick and Alan Wilson. Make a comparative analysis of the three concepts of the planning process. (15)
3. (a) Write down the development strategy of the present Government of Bangladesh. (05)
(b) Classify Projects as per the Planning Commission in Bangladesh. (05)
(c) Write down the structure and functions of National Economic Council (NEC) of the Planning Commission in Bangladesh. (12)
(d) Briefly describe the major conventional steps of Development Plan Preparation at city level in Bangladesh. (13)
4. (a) Write down your understanding about development plan with example. (05)
(b) Draw a schematic diagram of four stages plan for a city (a local scale plan) and explain with proper example. (10)
(c) "We need to distinguish three quite separate stages in the evolution of planning theory to understand the planning process" - Explain the statement (with proper example if it requires) (10)
(d) Why Master Plan concept is called Blue Print planning? What do you understand by Critical Balance in Strategic Planning? (10)



Section – B

5. (a) Narrate your concept on "Planning". "Involvement of grassroots community is a vital instrument for achieving sustainable development? – do you agree? Why? (08)
- (b) "A separate town planning profession gradually developed after the First World War" - Write about the initiatives contributed the most here. (07)
- (c) How will you define "Paradigm Shift" in planning theory? Use appropriate example to explain. (08)
- (d) Make comparative analysis between the following issues (any three): (12)
- (i) Positivism and Post-positivism
 - (ii) Spatial and Aspatial Planning
 - (iii) Conservation and Preservation
 - (iv) Top-down and Bottom-up approach of planning
6. (a) "Comprehensive planning is a rigid land use plan, which is difficult to apply in practical field"- Justify the statement with proper example. (10)
- (b) "Rational planning has a fixed sequence of tasks to solve any problem" – Explain with necessary example. (08)
- (c) In which situations "Incremental Planning" is most suitable to apply? Explain with appropriate example. (07)
- (d) "Transactive Planning is a process of mutual learning"- Identify the contribution of planner and client here. (10)
7. (a) "Understanding Target Audience" and "Developing Communication Strategy" are two major tasks of advocacy planning process. Explain with necessary example. (12)
- (b) Write about the initiatives that ensure "Adaptive learning and consensus building" in adaptive sustainable planning theory. (08)
- (c) Explain the concept "SITAR" in planning theory. Provide a short overview of each planning approaches used in SITAR model. (15)
8. (a) Make an overall comparison among the planning theories you have learnt on the ground "Public Participation". (15)
- (b) Write short notes on any four of the following issues: (20)
- (i) Negotiator
 - (ii) Inclusive city
 - (iii) Decentralization
 - (iv) Rigidness in planning
 - (v) Sustainable planning
 - (vi) Hard facts of planning



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2016
URP 1111
(History/Evolution of Human Settlements)

Full Marks: 210

Time: 3.0 hrs

- N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What were the major innovations of the Stone Age? How did these innovations change the lifestyle of the people? (15)
(b) Describe the following forms of human settlement: (10)
(i) Primitive organized
(ii) Dynamic urban settlement
(c) Illustrate the necessity of site factors for a settlement with appropriate example. (10)
2. (a) Write short notes on the following (04 × 05): (20)
(i) Conurbation
(ii) Mega City
(iii) Ring City
(iv) Central Business District
(b) Differentiate between urban and rural settlements using specific criteria. (15)
3. (a) Explain how factors like resources, transportation and government affect rural settlements. (15)
(b) Briefly describe the contents of a garden city. (20)
4. (a) How did Le Corbusier plan to reduce the decongestion of the center of the city? (15)
(b) Describe Linear City using Madrid as an example. (10)
(c) Write short notes on the following (03 × 05): (15)
(i) Vanga
(ii) Samatata
(iii) Harikela

Section – B

5. (a) Define human settlement. "Human settlement consists of both natural and man-made elements" – justify this statement by referring different elements of human settlements. (10)
(b) Describe the functions of a human settlement. Classify and explain different types of human settlements based on their functions. (10)
(c) Explain different patterns of settlement with necessary sketches. (09)
(d) Draw the hierarchy of human settlements based on population size. (06)



6. (a) 'The technology Roman used in construction methods is still used today'- (10)
explain the statement.
- (b) Discuss the main features of Mesopotamian civilization. (15)
- (c) What were the main features of 'City of Kahun' during the Egyptian (10)
civilization?
7. (a) Elucidate the main features of "Pont Du Gard Bridge" with sketches. (10)
- (b) Describe the salient features of 'Parthenon' with necessary sketches. (15)
- (c) Briefly discuss the city planning concept of 'Mohenjadaro' and 'Harappa' (10)
8. (a) Describe the evolutions in 'Transportation' and 'Health and Safety' during (10)
Industrial Revolution.
- (b) Discuss the impacts of Industrial Revolution on Human Settlements. (15)
- (c) Illustrate the general concepts of the Baroque city. (10)
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